

Objectives

- Know how to use Smith Chart and what is it useful for
- Use it to design single-stub matching network

Reflection Coefficient

$$\tilde{V}(z) = V_0^+ e^{-j\beta z} + V_0^- e^{j\beta z}, \quad (2.54a)$$

$$\tilde{I}(z) = \frac{V_0^+}{Z_0} e^{-j\beta z} - \frac{V_0^-}{Z_0} e^{j\beta z}. \quad (2.54b)$$

$$\text{load impedance } Z_L \quad Z_L = \frac{\tilde{V}_L}{\tilde{I}_L}. \quad (2.55)$$

$$\tilde{V}_L = \tilde{V}(z=0) = V_0^+ + V_0^-, \quad (2.56a)$$

$$\tilde{I}_L = \tilde{I}(z=0) = \frac{V_0^+}{Z_0} - \frac{V_0^-}{Z_0}. \quad (2.56b)$$

$$Z_L = \left(\frac{V_0^+ + V_0^-}{V_0^+ - V_0^-} \right) Z_0. \quad (2.57)$$

$$V_0^- = \left(\frac{Z_L - Z_0}{Z_L + Z_0} \right) V_0^+. \quad (2.58)$$

$$\begin{aligned} \Gamma &= \frac{V_0^-}{V_0^+} = \frac{Z_L - Z_0}{Z_L + Z_0} \\ &= \frac{Z_L/Z_0 - 1}{Z_L/Z_0 + 1} \\ &= \frac{z_L - 1}{z_L + 1} \quad (\text{dimensionless}), \end{aligned} \quad (2.59)$$

$$z_L = \frac{Z_L}{Z_0} \quad \text{normalized load impedance.}$$

► The ratio of the amplitudes of the reflected and incident voltage waves at the load is known as the **voltage reflection coefficient** Γ . ◀

Incident wave

Reflected wave

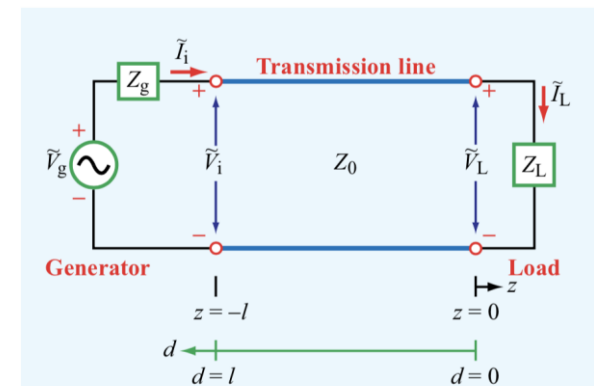
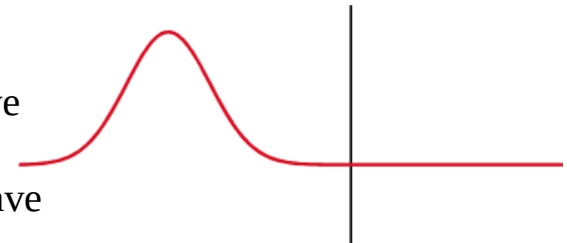


Figure 2-12 Transmission line of length l connected on one end to a generator circuit and on the other end to a load Z_L . The load is located at $z = 0$ and the generator terminals are at $z = -l$. Coordinate d is defined as $d = -z$.

$$\Gamma = |\Gamma| e^{j\theta_r}, \quad (2.62)$$

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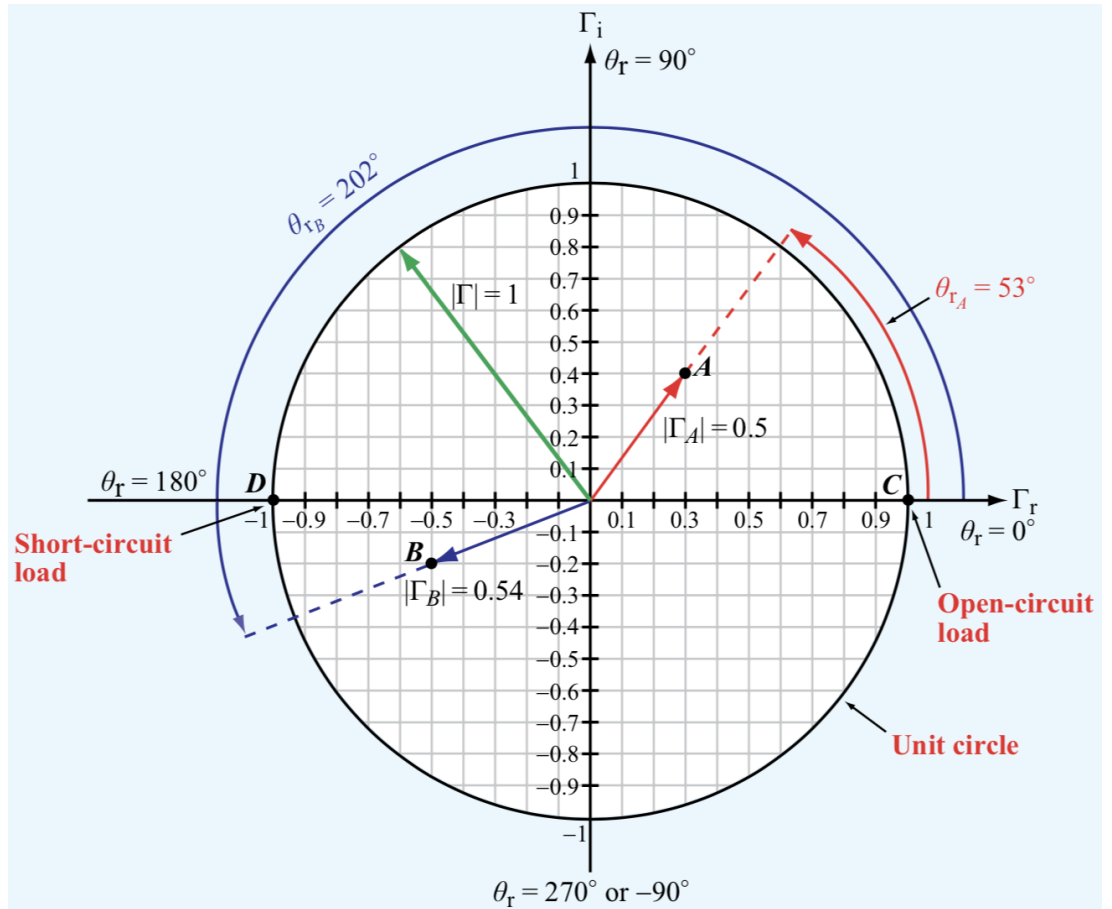
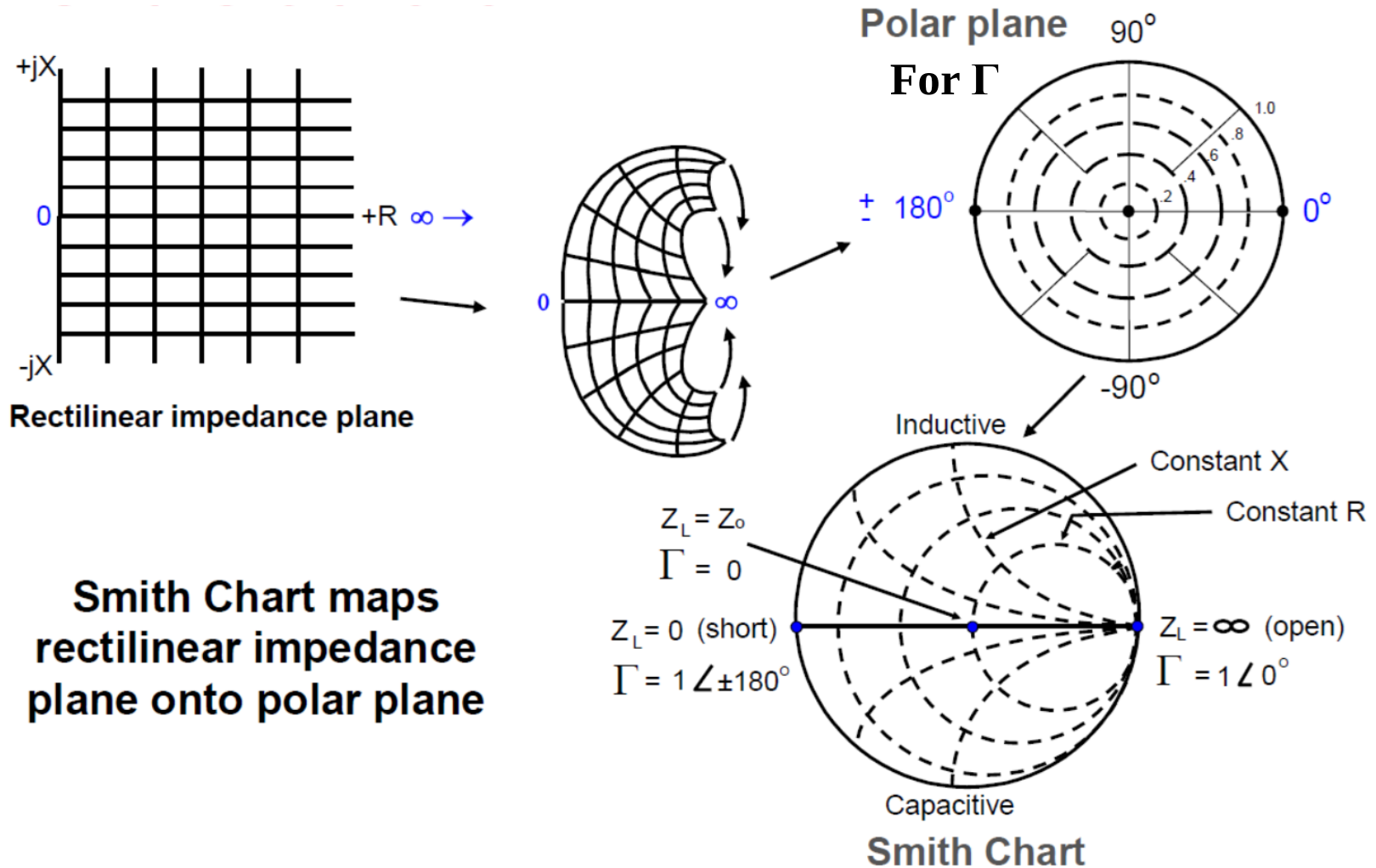


Figure 2-24 The complex Γ plane. Point A is at $\Gamma_A = 0.3 + j0.4 = 0.5e^{j53^\circ}$, and point B is at $\Gamma_B = -0.5 - j0.2 = 0.54e^{j202^\circ}$. The unit circle corresponds to $|\Gamma| = 1$. At point C, $\Gamma = 1$, corresponding to an open-circuit load, and at point D, $\Gamma = -1$, corresponding to a short circuit.

What is the Smith chart?



How Impedance align with the gamma polar plot? (No need to know this)

$$\Gamma = \frac{Z_L/Z_0 - 1}{Z_L/Z_0 + 1} = \frac{z_L - 1}{z_L + 1}$$

$$z_L = \frac{1 + \Gamma}{1 - \Gamma} \quad (2.112)$$

$$z_L = r_L + jx_L$$

$$r_L = \frac{1 - \Gamma_r^2 - \Gamma_i^2}{(1 - \Gamma_r)^2 + \Gamma_i^2}$$

$$x_L = \frac{2\Gamma_i}{(1 - \Gamma_r)^2 + \Gamma_i^2}$$

Equation for a circle

$$\left(\Gamma_r - \frac{r_L}{1 + r_L}\right)^2 + \Gamma_i^2 = \left(\frac{1}{1 + r_L}\right)^2 \quad (2.116)$$

r_L circles

r_L circles are contained inside the unit circle

The standard equation for a circle in the x - y plane with center at (x_0, y_0) and radius a is

$$(x - x_0)^2 + (y - y_0)^2 = a^2 \quad (2.117)$$

A similar manipulation of the expression for x_L given by Eq. (2.115b) leads to

$$(\Gamma_r - 1)^2 + \left(\Gamma_i - \frac{1}{x_L}\right)^2 = \left(\frac{1}{x_L}\right)^2 \quad (2.118)$$

x_L circles

Only parts of the x_L circles are contained within the unit circle

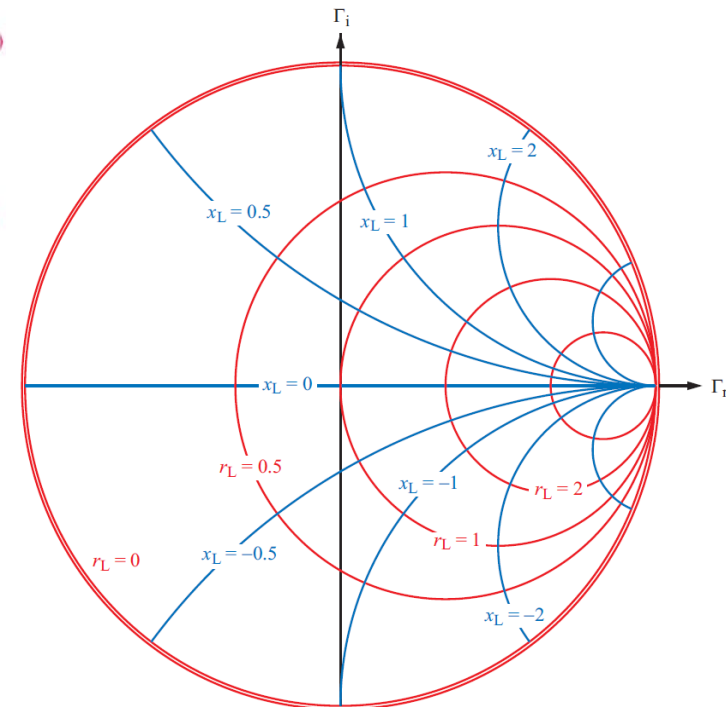


Figure 2-25: Families of r_L and x_L circles within the domain $|\Gamma| \leq 1$.

What is the Smith chart?

- It's really just a plot of complex reflection overlaid with an impedance and/or admittance grid referenced to a 1-ohm characteristic impedance.
- The Smith chart contains almost all possible impedances, real or imaginary, within one circle.
- It is possible to go **outside the Smith chart "unity" circle**, but only with an **active device** (because this implies $-ve$ resistance).
- Developed in 1939 by P. W. Smith as a graphical tool to analyze and design transmission-line circuits
- Today, it is used to characterize the performance of microwave circuits

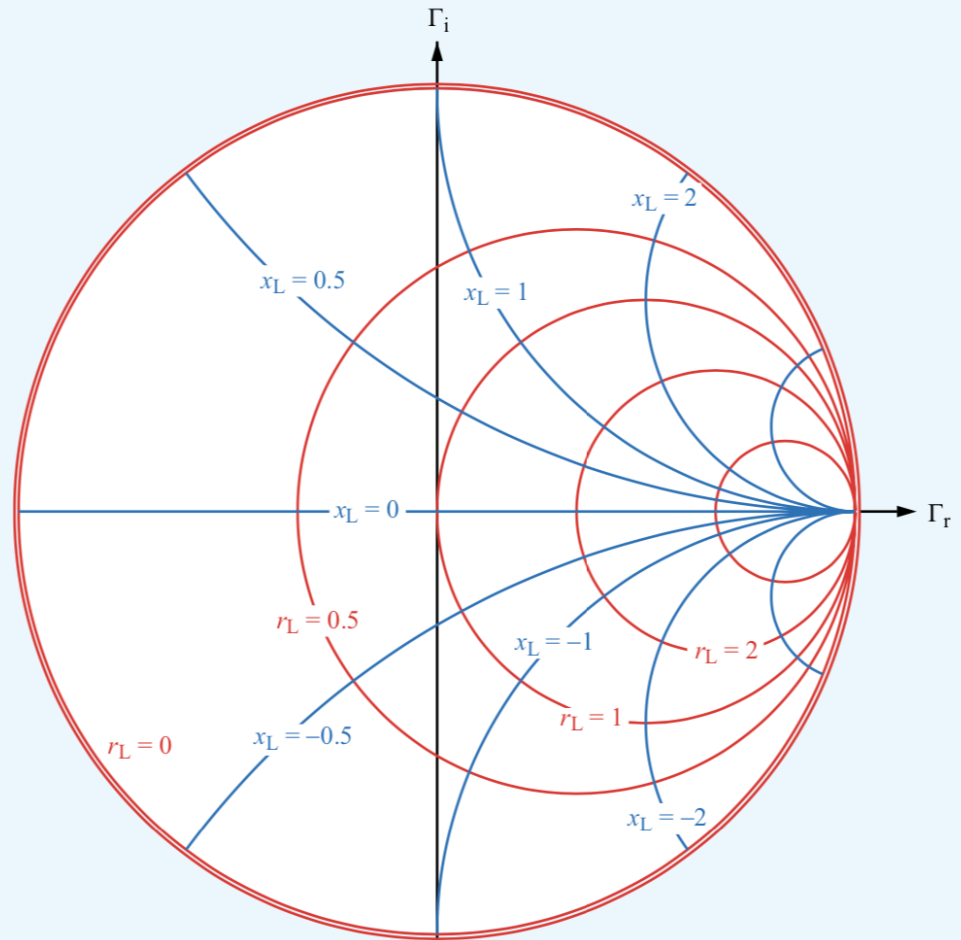


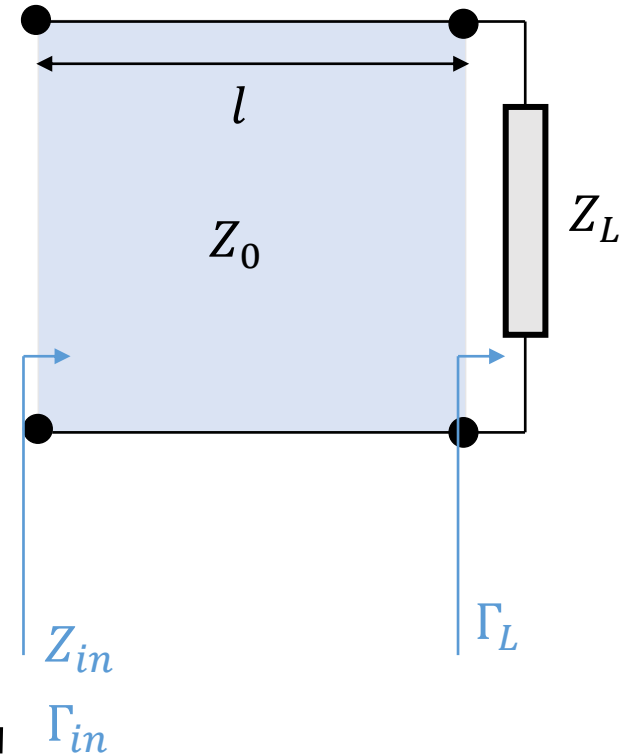
Figure 2-25 Families of r_I and x_I , circles within the domain $|\Gamma| \leq 1$.

Alternative ways to deal TL

- $\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}$
- $Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan(\beta l)}{Z_0 + jZ_L \tan(\beta l)}$
- $SWR = \frac{1+|\Gamma|}{1-|\Gamma|}$ & $RL = -20 \log |\Gamma|$

Few tips :

- $|\Gamma_{in}| = |\Gamma_L|$ but $\Gamma_{in} \neq \Gamma_L$
- $Z_{in} = Z_0$ (for matched load) NOT function of l !
- $Z_{in} = jZ_0 \tan(\beta l)$ (for short-circuited load)
- $Z_{in} = -jZ_0 \cot(\beta l)$ (for open-circuited load)



What is the Smith chart useful for?

- Getting Z from Γ and vice versa
- Getting VSWR
- Getting Z_{in} of a TL with length l terminated by Z_L
- Getting Impedance $\Leftrightarrow$ Admittance Conversion: $Z \Leftrightarrow Y = \frac{1}{Z}$
- Designing single-stub matching network (later)

Secret #1: Knowing Z = Knowing Γ

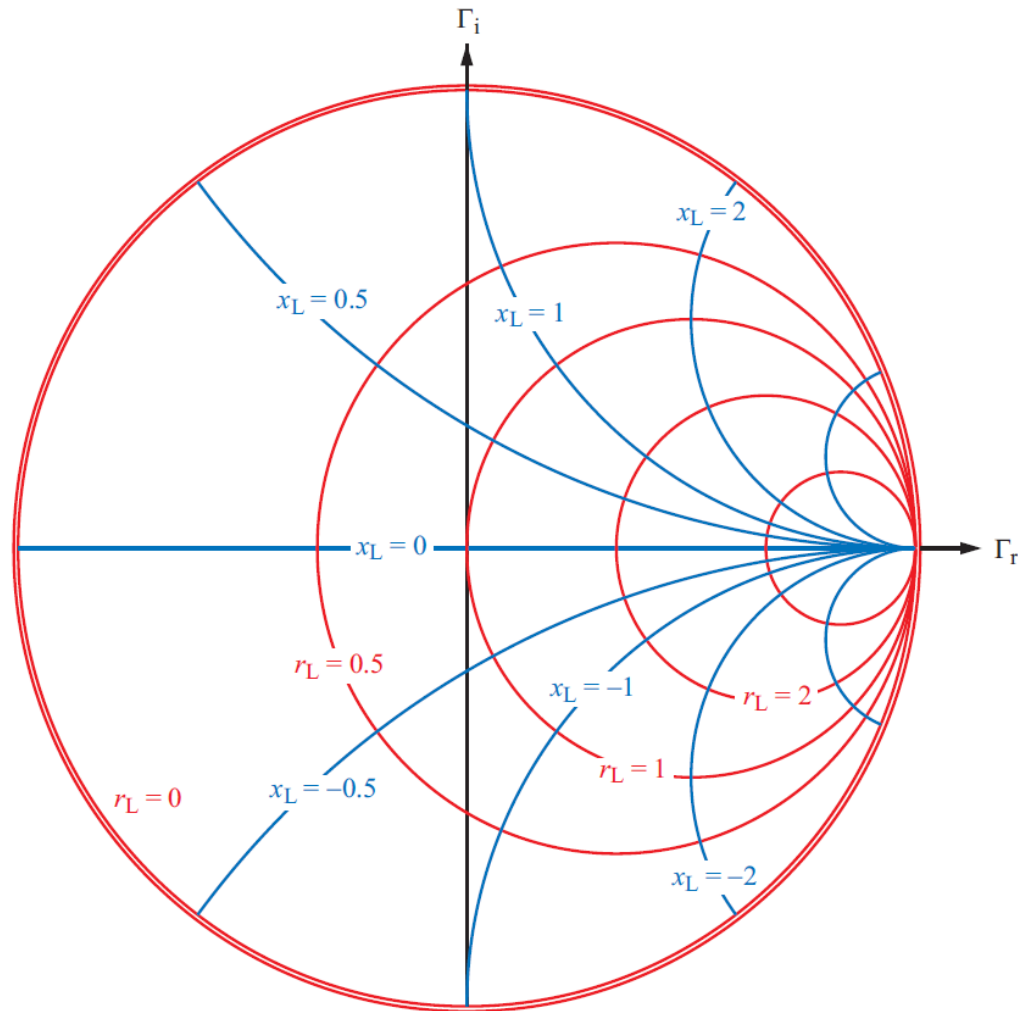
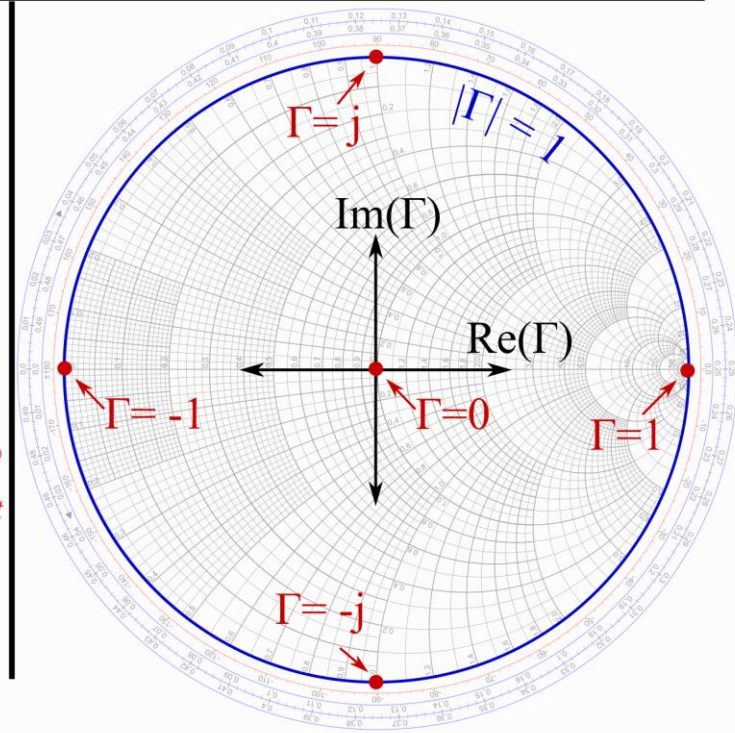
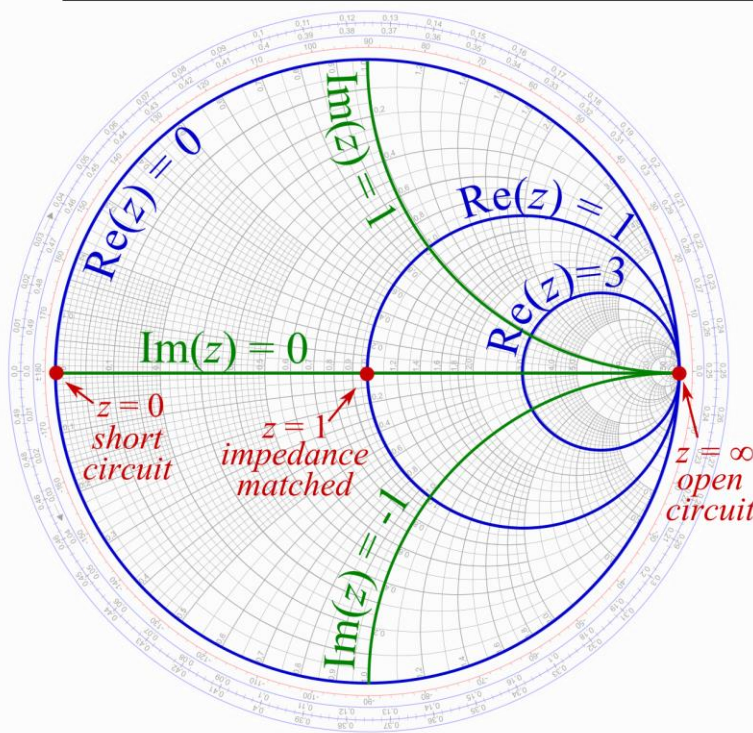


Figure 2-25: Families of r_L and x_L circles within the domain $|\Gamma| \leq 1$.

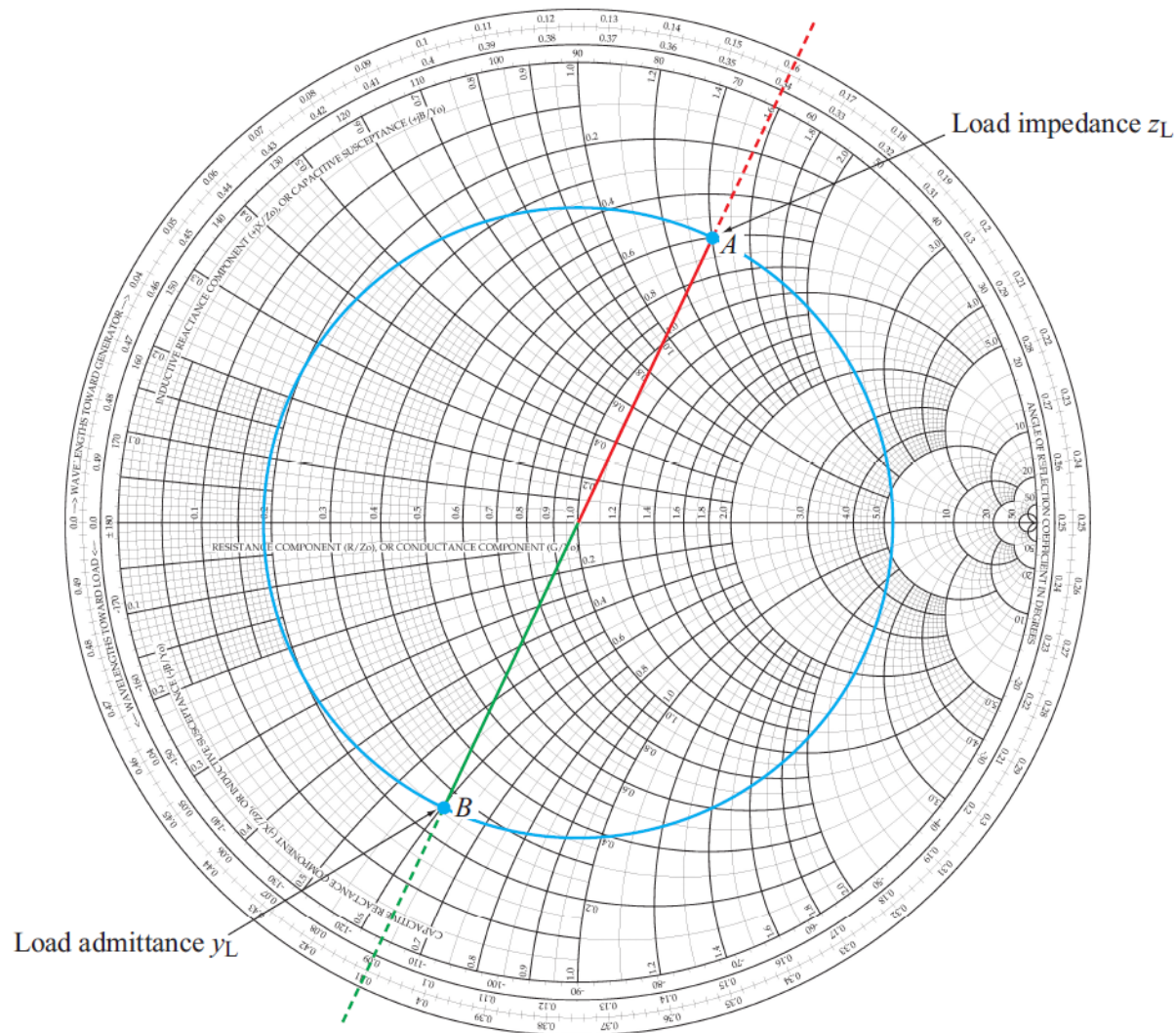
A diagram of a transmission line with characteristic impedance Z_0 connected to a load impedance Z_L . The incident voltage wave is $V_{incident}$ and the reflected voltage wave is $V_{reflected}$.

$$z = \frac{Z_L}{Z_0}$$

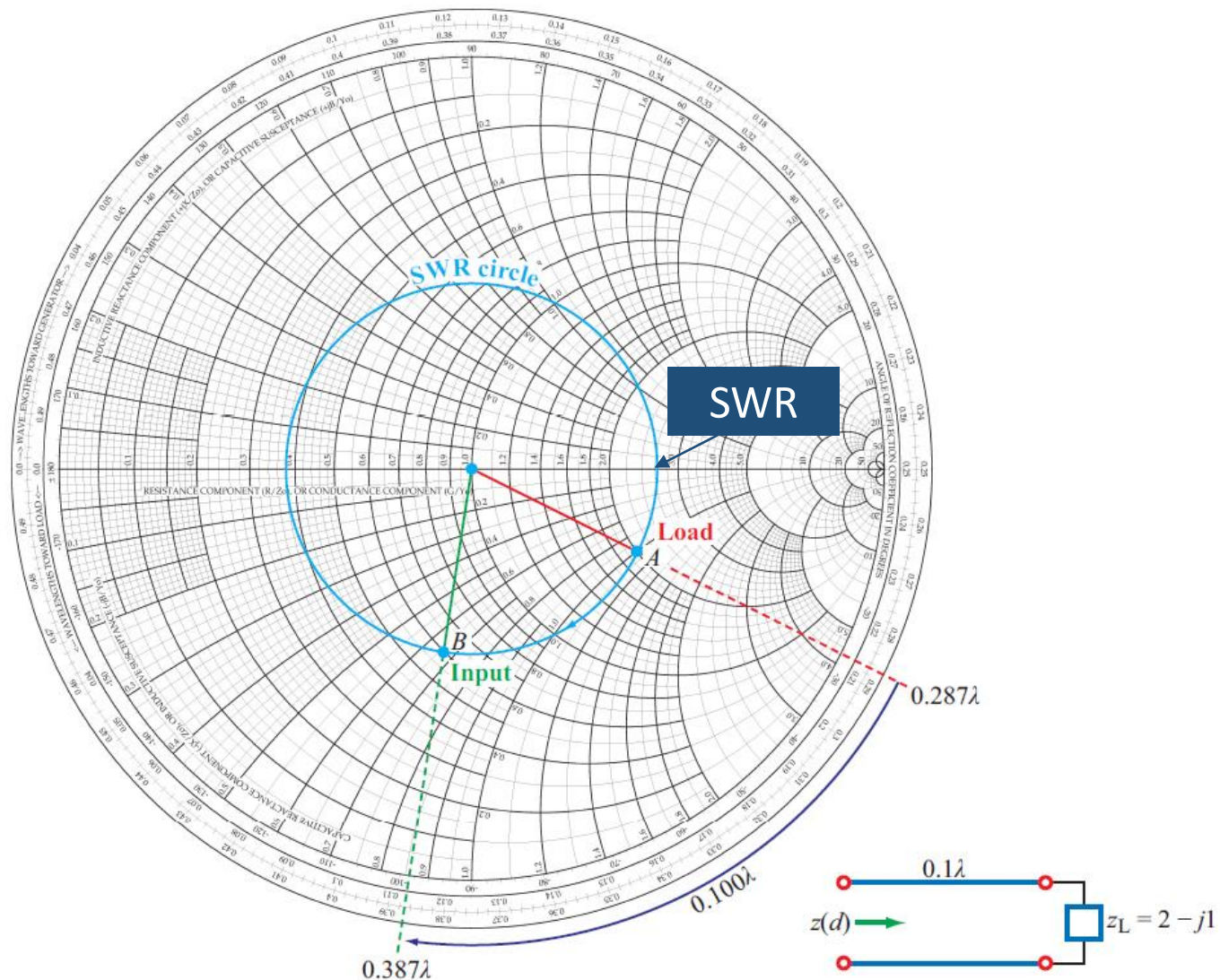
$$\Gamma = \frac{V_{reflected}}{V_{incident}}$$



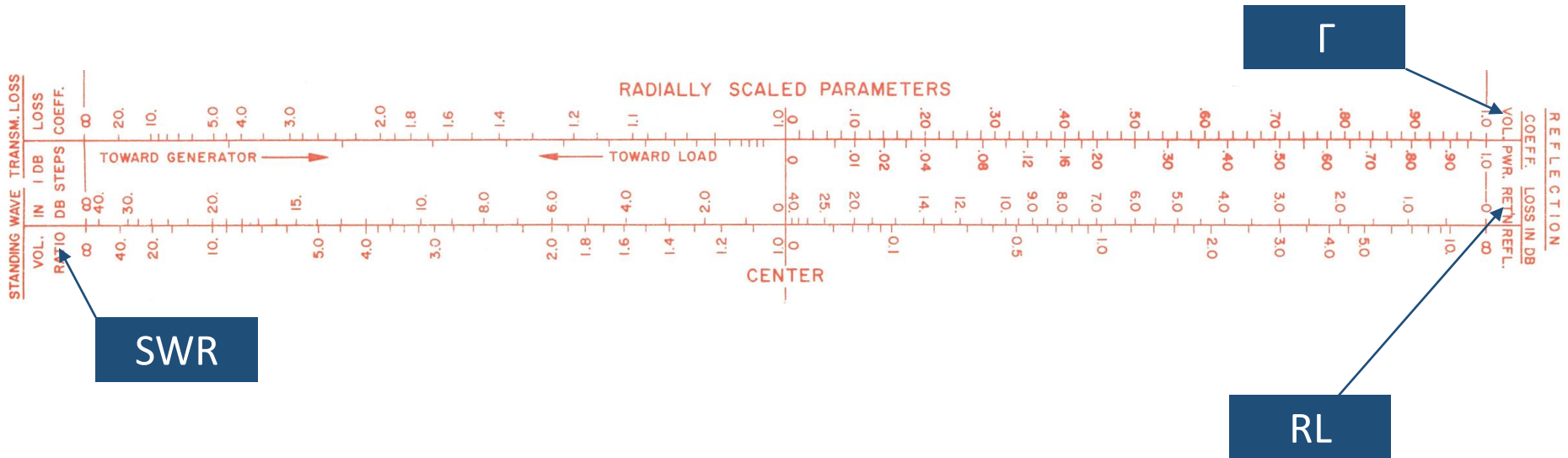
Secret #2: Go through Center for Impedance $\Leftrightarrow$ Admittance Conversion



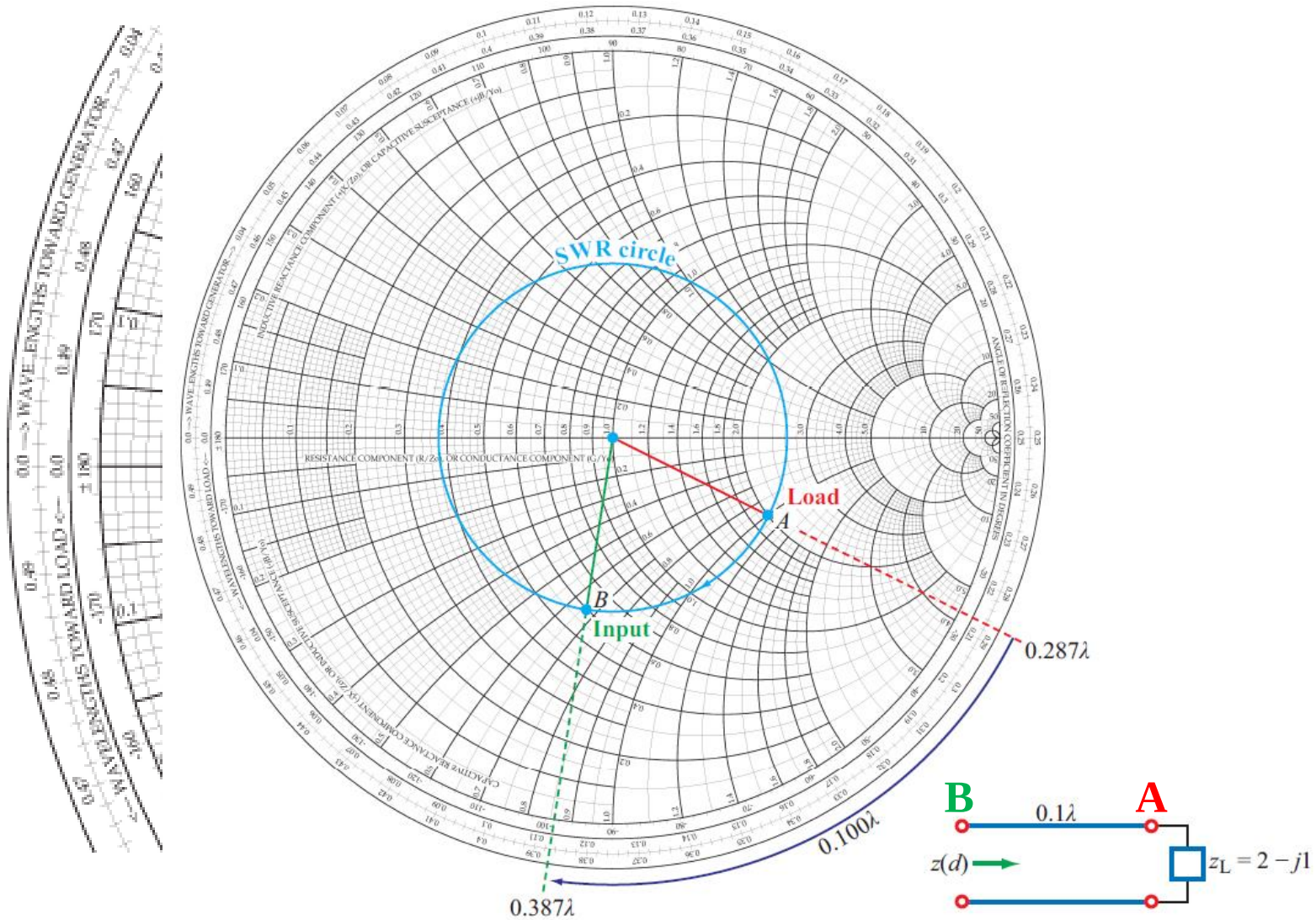
Secret # 3: Plot a circle know the SWR



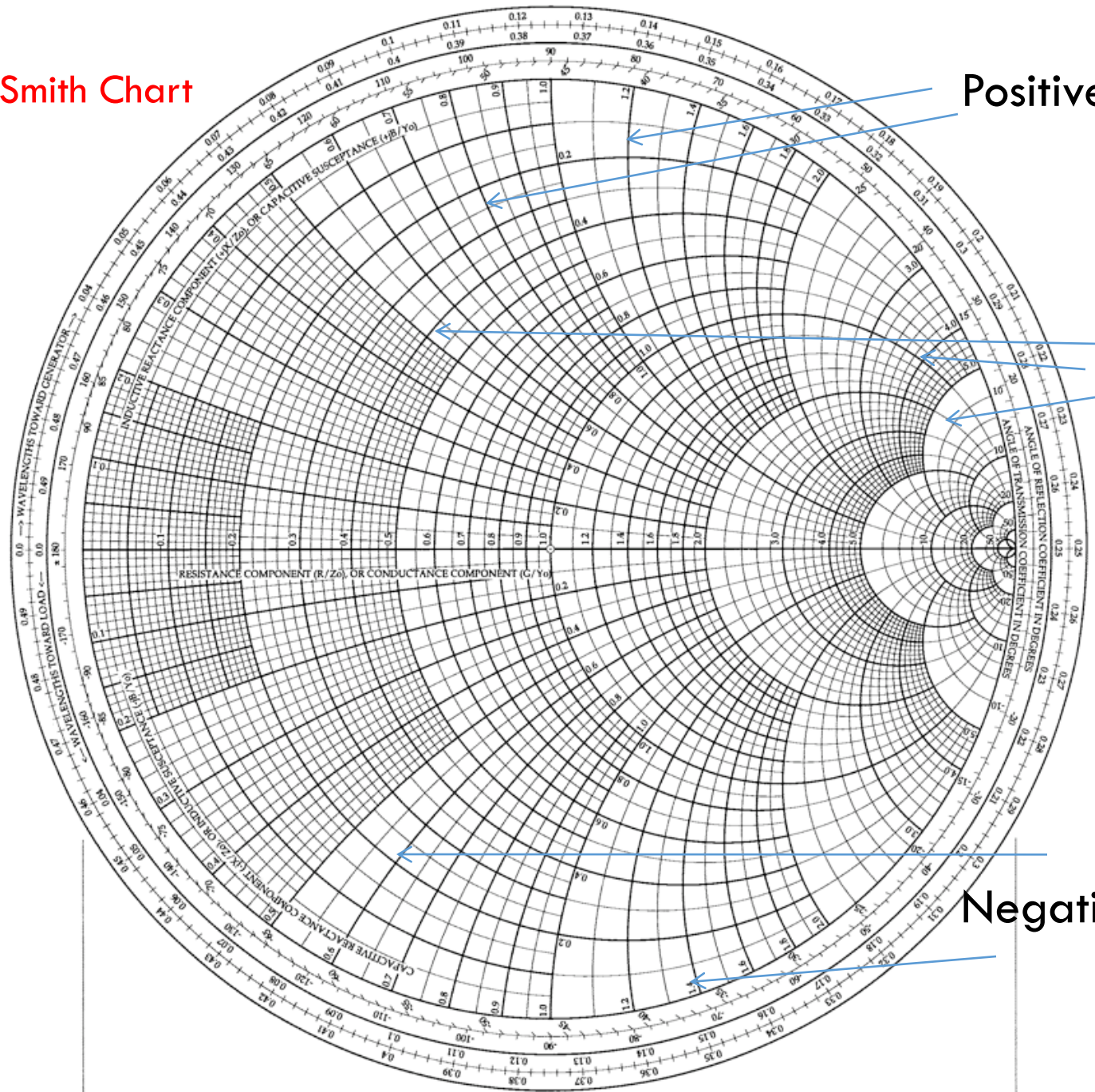
Secret # 3: Plot a circle know the SWR



Secret # 4: Know Z_{in} by rotating Z_L along SWR circle



Complete Smith Chart



Positive x_L Circles

r_L Circles

Negative x_L Circles

Find Γ_L and VSWR for $Z_L = 25 + j25$

- Method 1: Using Equations (Not Smith chart)

$$Z_n = \frac{25}{50} + j\frac{25}{50} = 0.5 + j0.5$$

We'll solve for the reflection coefficient and VSWR by hand.

Recall, the reflection coefficient in terms of the normalized impedance is:

$$\Gamma = \frac{V^-}{V^+} = \frac{Z - Z_o}{Z + Z_o} = \frac{Z_n - 1}{Z_n + 1}$$

Therefore, the normalized reflection coefficient for the normalized impedance is given as:

$$\Gamma = \frac{0.5 + j0.5 - 1}{0.5 + j0.5 + 1} = \frac{-0.5 + j0.5}{1.5 + j0.5} = \frac{-0.5 + j1}{2.50} = -0.2 + j0.4$$

$$|\Gamma| = \sqrt{0.2^2 + 0.4^2} = \sqrt{0.20} = 0.447$$

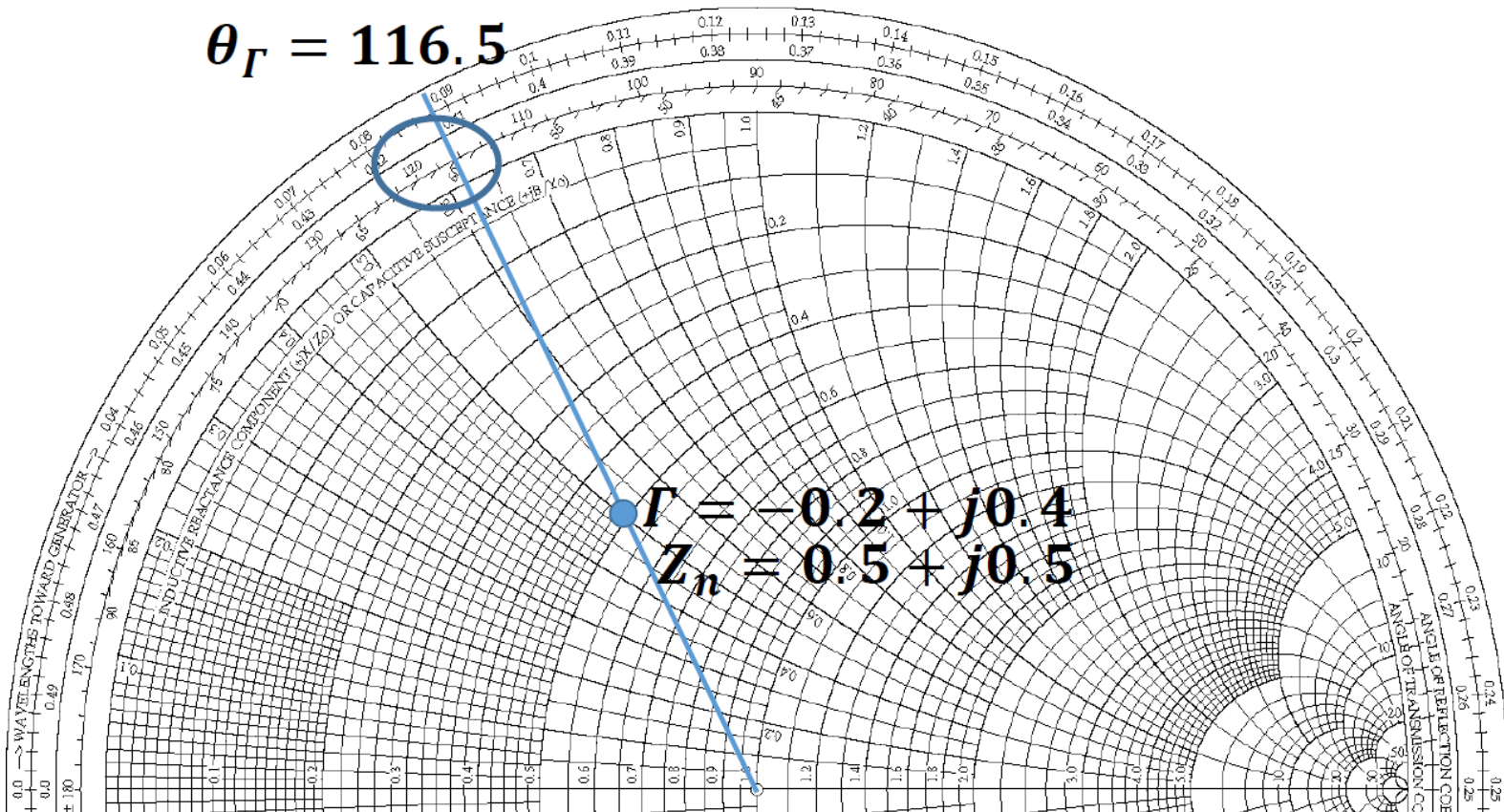
The VSWR for the normalized impedance is given as:

$$VSWR = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1 + 0.447}{1 - 0.447} = \frac{1.447}{0.553} = 2.62$$

Find Γ_L and VSWR for $Z_L = 25 + j25$

- Method 2: Using Smith chart

$$Z_n = \frac{25}{50} + j \frac{25}{50} = 0.5 + j0.5$$



Find Γ_L and VSWR for $Z_L = 25 + j25$

- Method 2: Using Smith chart

1.) Normalize the load impedance and plot the normalized load impedance.

$$Z_n = \frac{25}{50} + j\frac{25}{50} = 0.5 + j0.5$$

2.) Draw a line from the center of the Smith chart to the edge of the chart, while passing through the normalized load impedance. The angle of the reflection coefficient at that point is

$$\theta_r = \tan^{-1} \left(\frac{\Gamma_i}{\Gamma_r} \right) = \tan^{-1} \left(\frac{0.40}{-0.20} \right) = -63.43^\circ \text{ or } 116.57^\circ$$

3.) Using a compass, place the compass needle at the center of the Smith chart and the other leg at the normalized load impedance. Make sure the distance is accurate.

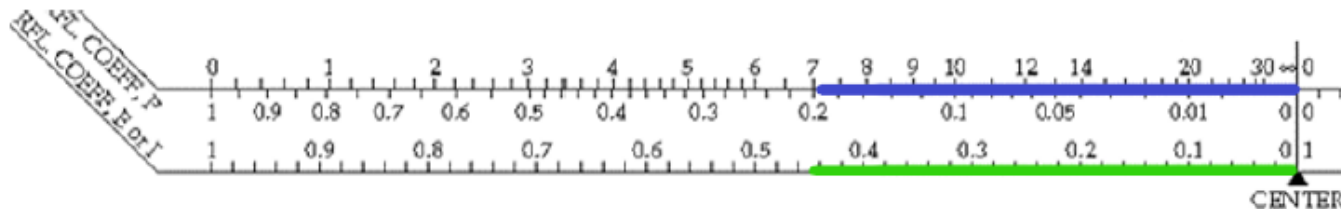
4.) Look at the scales below the Smith chart and find the one labeled “RFL COEFF E, OR I”.

5.) Place the compass needle on the scale at zero and then, use the other leg to mark the scale. Remove the compass and read the number at the mark.

Find Γ_L and VSWR for $Z_L = 25 + j25$

- Method 2: Using Smith chart

6.) The absolute value of the voltage reflection coefficient is equal to 0.447 as seen by the green line. The absolute value of the power reflection coefficient is equal to 0.199 as seen by the blue line.



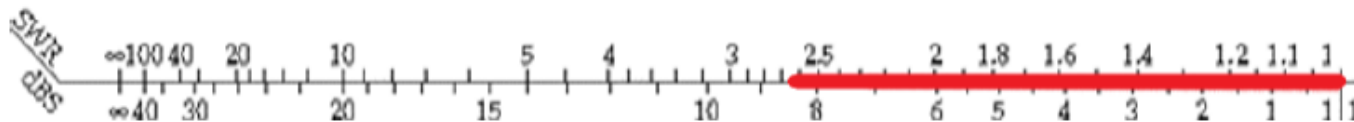
7.) The voltage reflection coefficient is:

$$\Gamma_E = 0.447 \angle 116.57^\circ = -0.2 + j0.4$$

8.) Find the scale find the named "SWR".

9.) Place the compass needle at one on that scale and then, use the other leg to mark the scale. Remove the compass and read the number at the mark.

10.) The VSWR is equal to 2.62 as seen by the red line.



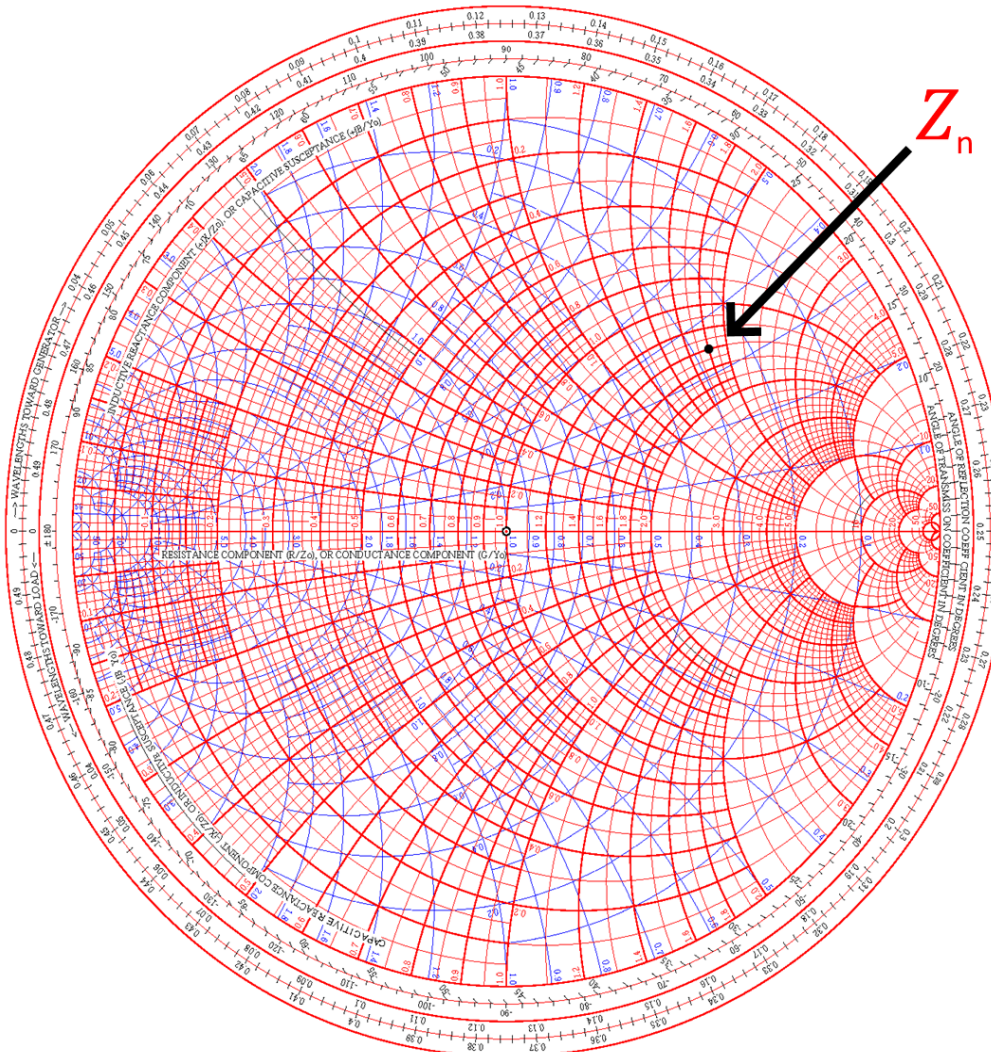
Example 2

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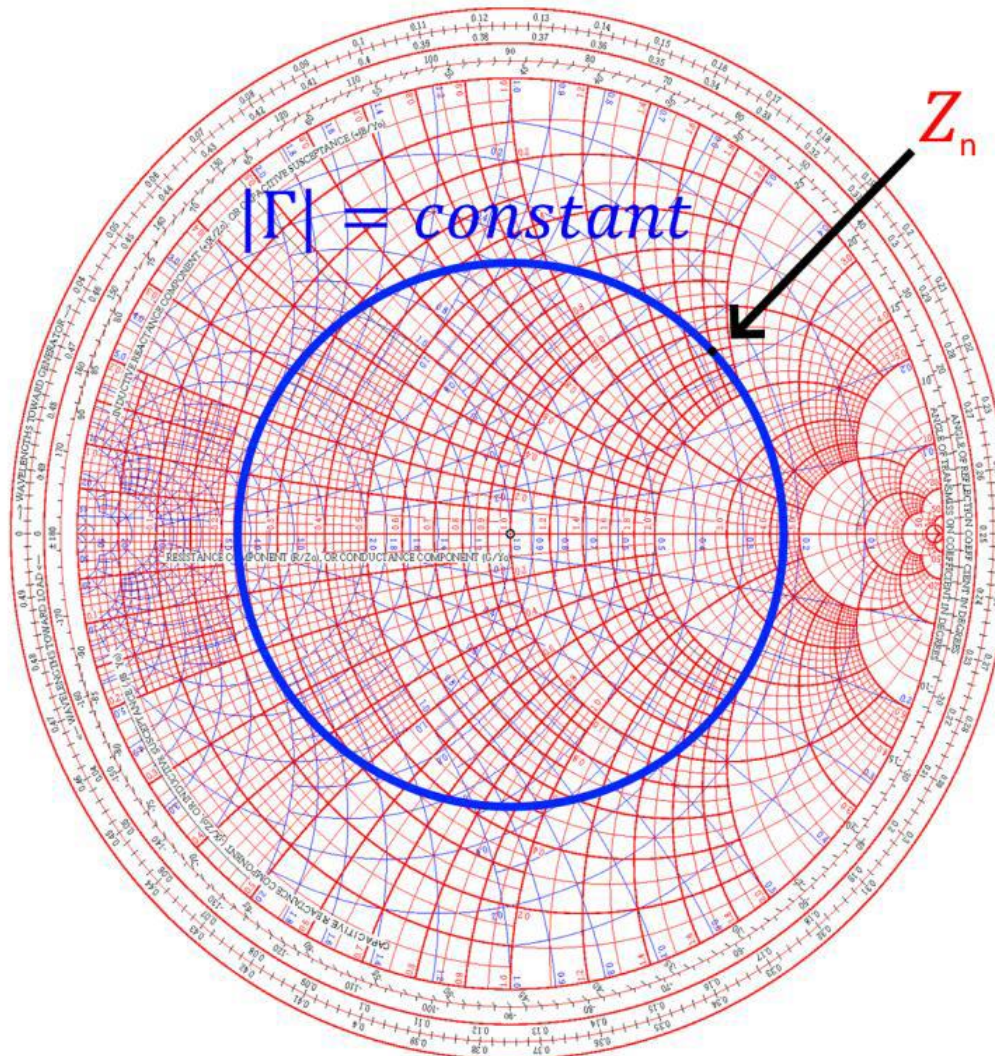
1) Plot the normalized load impedance $Z_n = \frac{Z_L}{Z_o} = 1.4 + j1.8$



Example 2

Suppose you wish to calculate the input impedance at a distance $\ell=0.1\lambda$ from the load, where $Z_o = 50\Omega$ and $Z_L = 70 + j90$

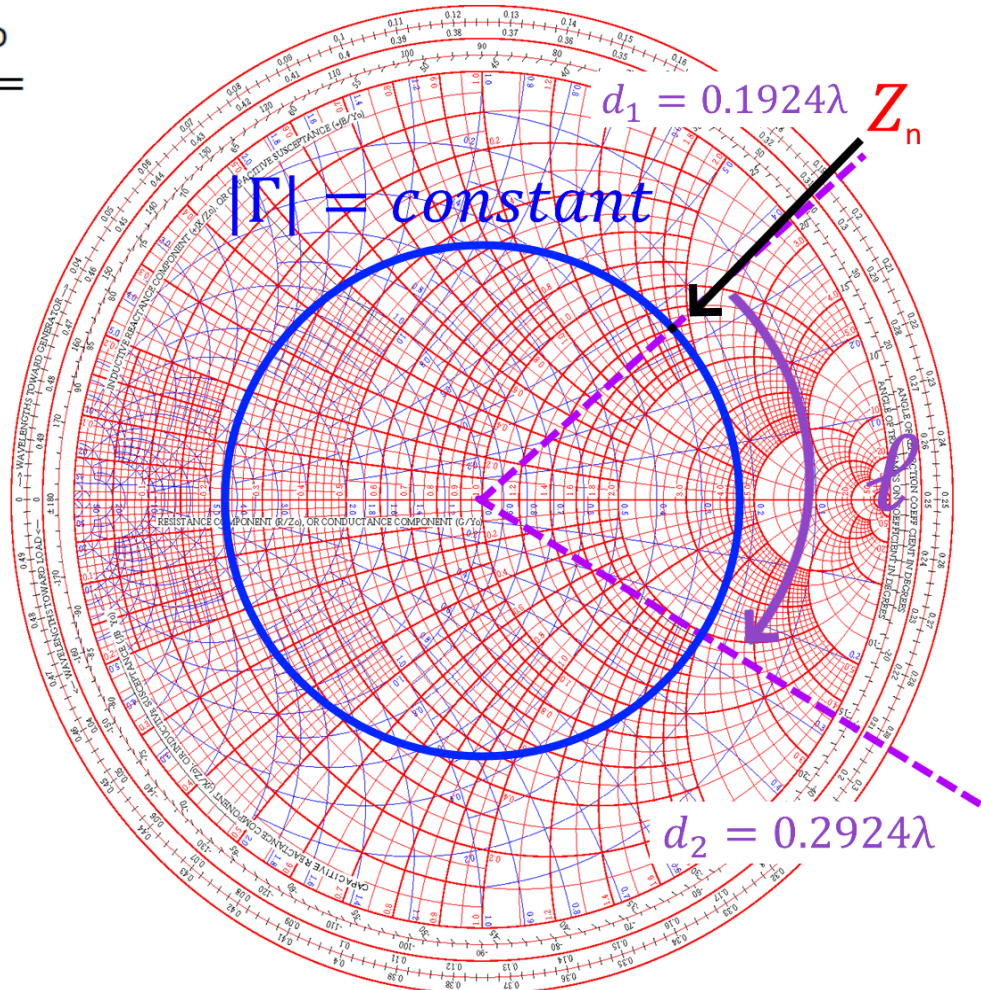
2) Draw the constant- $|\Gamma|$ circle that passes through the plotted normalized impedance



Example 2

Suppose you wish to calculate the input impedance at a distance $\ell = 0.1\lambda$ from the load, where $Z_o = 50\Omega$ and $Z_L = 70 + j90$

3) Rotate the impedance clockwise around the constant- $|\Gamma|$ circle by a distance of 0.1λ on the wavelengths-toward generator scale (Note: the initial value on this scale is $d_1 = 0.1924\lambda$, so rotating by 0.1λ will take us to $d_2 = d_1 + \ell = 0.2924\lambda$)



4) Read the new impedance off the chart.
In this case, $Z_{in,normalized} = 1.89 - j1.95$

so, $Z_{in} = 94.7 - j97.5\Omega$

Example 3

$$Z_L = 25 - j50$$

Find:

- a) Γ_L
- b) Y_L
- c) SWR
- d) RL
- e) Z_{in} after 0.25λ
- f) Γ_{in} after 0.25λ

